Emily A. O'Rourke

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EDUCATION:

University of Pennsylvania Perelman School of Medicine — Philadelphia, PA
PhD Candidate, Cell and Molecular Biology — 2021 – present
Subgroup: Microbiology, Virology, and Parasitology

Massachusetts Institute of Technology — Cambridge, MA
Bachelor of Science in Biology — 2017 – 2021
Concentration in Anthropology

Franklin High School — El Paso, TX
Valedictorian — 2014 – 2017

RESEARCH EXPERIENCE:

University of Pennsylvania Department of Microbiology — Philadelphia, PA
Ph.D. Thesis Research Lab — July 2022 – present
Advisor: Sunny Shin, Ph.D.

- Thesis title: Inflammasome-driven cell death in human macrophages during *Salmonella* infection
- Utilized molecular-biology wet lab techniques such as tissue-culture, western blotting, ELISA, and cell death assays to probe the mechanisms underlying programmed cell death in response to *Salmonella* infection

University of Pennsylvania Department of Microbiology — Philadelphia, PA
Cell and Molecular Biology Research Rotations — September 2021 – June 2022
Advisors: (1) Sunny Shin, Ph.D., (2) Audrey Odom John, MD, Ph.D., (3) Joseph W. St. Geme, MD

(1) Inflammasome-mediated control of *Salmonella typhimurium* infection in human macrophages
(2) HAD2 function in isoprenoid biosynthesis in *Plasmodium falciparum*
(3) The role of individual *PAM* genes in *Kingella kingae* LPS biosynthesis

Massachusetts Institute of Technology Department of Biology — Cambridge, MA
Undergraduate Research Opportunities Program (UROP) — 2018 – 2020
Advisor: Rebecca Lamason, Ph.D.

- Investigated how the *Rickettsia parkeri* effector protein Sca4 interacts with host cell processes to promote infection. Focused on Sca4 subcellular localization, interacting protein domains, and mode of secretion

University of Pennsylvania Department of Microbiology — Philadelphia, PA
Summer Undergraduate Internship Program (SUIP) — May – August 2020
Advisor: Kellie Ann Jurado, Ph.D.

- Examined the mechanisms used by SARS-CoV-2 to evade, bypass, and subvert the innate immune system in humans
- Composed a final research proposal to delineate the molecular mechanism of SARS-CoV-2 immune evasion by Nsp13

Università Iuav di Venezia & Massachusetts Institute of Technology Department of Civil Engineering — Venice, Italy

MIT International Science and Technology Initiatives (MISTI) May – August 2018
Advisors: Andrew Whittle, Ph.D. (MIT), Massimiliano Scarpa, Ph.D. (IUAV)

- Investigated and synthesized current scientific literature regarding biogas digesters as a potential source of renewable energy in Northern Italy
- Wrote a cumulative review highlighting the benefits of different options for biogas digesters

PUBLICATIONS:

Marisa S. Egan, **Emily A. O'Rourke**, Shrawan Kumar Mageswaran, Biao Zuo, Inna Martynyuk, Tabitha Demissie, Emma N. Hunter, Antonia R. Bass, Yi-Wei Chang, Igor E. Brodsky, Sunny Shin (2025) Inflammasomes primarily restrict cytosolic Salmonella replication within human macrophages *eLife* 12:RP90107 — March 2025

AWARDS & HONORS:

American Heart Association Predoctoral Fellowship — 2024 – present
Awarded 2 years of research funding to enhance the integrated research and clinical training of promising students who are matriculated in pre-doctoral training programs and who intend careers as scientists

HHMI Gilliam Predoctoral Fellowship – Finalist — 2024

Cell and Molecular Biology NIH T32 Training Grant — 2022 – 2024
Selected for 2 years of research funding and various training experiences.

Ford Foundation Predoctoral Fellowship – Honorable Mention — 2023

Fontaine Society Fellowship — 2021 – present
Awarded to support the academic development of University of Pennsylvania PhD students whose research and experiences will enhance the diversity of the professoriate

Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) Scholarship — 2020
Award to cover costs to attend the 2020 SACNAS Conference

Johnson & Johnson UROP Scholars Grant — 2019
Selected for an undergraduate summer research program for MIT women conducting STEM research. Awarded funding for a full-time independent summer research project

MIT UROP Research Grant — 2017 - 2020
Awarded funding for 5 semesters of independent research based on a research proposal

PRESENTATIONS:

Cold Spring Harbor Asia: Bacterial Infection & Host Defense — May 2026
Invited oral presentation, Suzhou, China

Philadelphia Infection & Immunity Forum PIIF — Dec 2023
Poster presentation, Philadelphia, PA

Cold Spring Harbor: Microbial Pathogenesis and Host Response — Sept 2023
Poster presentation, Cold Spring Harbor, NY

26th Annual Cell and Molecular Biology Symposium — Oct 2023

Poster presentation, Philadelphia, PA

Microbiology, Virology, and Parasitology Annual Retreat Aug 2023
Poster presentation, Swarthmore, PA

Cell and Molecular Biology End of Rotation Presentations 2021 – 2022
Summarized 10 weeks of research at lab group meeting

SUIP Program Final Presentation 2020
Gave a formal ten-minute presentation communicating my final research proposal

Johnson & Johnson UROP Scholars Poster Presentation 2019
Presented a poster titled "Investigating and characterizing Sca4 secretion in *Rickettsia parkeri*"

TEACHING, MENTORSHIP, & OUTREACH:

Research Lab mentor for UPenn Vagelos Scholar Undergraduate Student Philadelphia, PA
2022 - 2026
- Mentored Talia Smith part time in lab during the semester, and full time during the summer in an independent research project.
- She presented a poster at the summer Vagelos Scholars Poster Symposium

SPAN 800: Spanish Conversation, *Course director and lead instructor* Philadelphia, PA
2023 - 2026
- Independently designed and lead a 0.5 credit course in the University of Pennsylvania Spanish Department

Residential Advisor: Gregory College House Philadelphia, PA
2023 - present
- Serve as a residential advisor in a University of Pennsylvania undergraduate dorm. Mentor a floor of 15 students.

MVP Recruitment, *Co-coordinator* Philadelphia, PA
Feb 2024
- Coordinated all MVP recruitment activities for prospective students during three visit weekends.

Science Education Academy (SEA), *Volunteer* Philadelphia, PA
2021 - 2022
- Led a free hands-on science activity for 5th grade students in West Philadelphia
- STEM non-profit (501c3) with the aim to support elementary grade students in their science education with hands-on exposure with scientific method and inquiry

MIT Biology Tutoring Program Cambridge, MA
2019 – 2021
- Tutored six MIT undergraduate students in two classes: Introductory Biology and Cell Biology
- Met weekly to clarify concepts and work through relevant problems

MIT Global Teaching Labs*, Student teacher* Haifa, Israel
January 2019
- Designed a week-long STEM course curriculum to lead hands-on activities for teaching fundamental principles of App Inventor to middle school students
- Taught four week-long workshops at four different schools in the ORT public school system to approximately 400 students total

ACTIVITIES AND SERVICE:

UPenn MicroIDEAs, General member 2022 – present
Inclusion, Diversity, Equity, Antiracism in the Sciences

UPenn SACNAS Chapter, General member 2021 – present
Society for Advancement of Chicanos/Hispanics and Native Americans in Science

UPenn CAMB Student Recruitment, Coordinating committee volunteer 2022
Planned and coordinated recreational and academic activities for the virtual CAMB interview weekends and in-person accepted students' weekends

MIT Mujeres Latinas, Secretary 2020 – 2021
Organized events to promote social and professional empowerment of Latinas in the MIT community.

MIT SUBE, Society of Underrepresented Biologists and Biological Engineers Aug. 2020 - 2021

MIT BUSA, Biology Undergraduate Student Association 2018 - 2021

ADDITIONAL EXPERIENCE:

Public Health Certificate Program, University of Pennsylvania Philadelphia, PA
Director: Hillary Nelson, Ph.D. 2021 - 2026

- Took "*Intro to Epidemiology*" and "*Foundations of Global Health*" courses
- Attended biweekly seminars on various public health topics

MIT HST.434 Evolution of an Epidemic, Study Abroad Durban, South Africa
Professors: Bruce Walker, M.D., Howard Heller, M.D. January 2020

- Selected to participate in a two-week class in South Africa sponsored by the Harvard-MIT Program in Health Sciences and Technology
- Learned about the evolution of the HIV epidemic and the various biosocial factors that contribute to disease spread

SKILLS:

Languages: Spanish (Native Speaker), Mandarin (Novice)

Computer: Basic R, Basic Python, Basic ImageJ, Adobe Illustrator, Adobe Photoshop, Adobe InDesign, Microsoft Office

Other interests: Graphic design, reading, distance running, and guitar